Techfest 2026-27
Meshmerize

Objective:

Design and build an autonomous robot capable of detecting and following a white line while tracking its direction of travel through a maze. The bot must first analyse the arena during a timed dry run, then use the path it has learned to navigate from the starting point to the ending point of the maze in the shortest possible time on its actual run.

Arena Design & Specifications:

The game field consists of an arena having dimensions 230 cm X 230 cm (L x B). It consists of the following:

1. The arena is composed of random paths made up of white Vinyl strips.
2. All the distances are shown in Fig. 1 and Fig. 2.
3. The Angle between two adjacent white lines in the path is 90°.
4. The width of all white stripes will be 30mm.
5. The figure below shows the sample arena. The actual arena at the competition will consist of alterations in the path.
6. A white box of 400 mm x 400 mm is present at the end zone of the arena to indicate the end position.

NOTE: The dimensions of the arena will be accurate to within 5% or 20 mm, whichever is less.

Checkpoints:

The checkpoints are shown in the figure below. There will be no circles on the arena; they are used as references
Each checkpoint carries **25 points.**

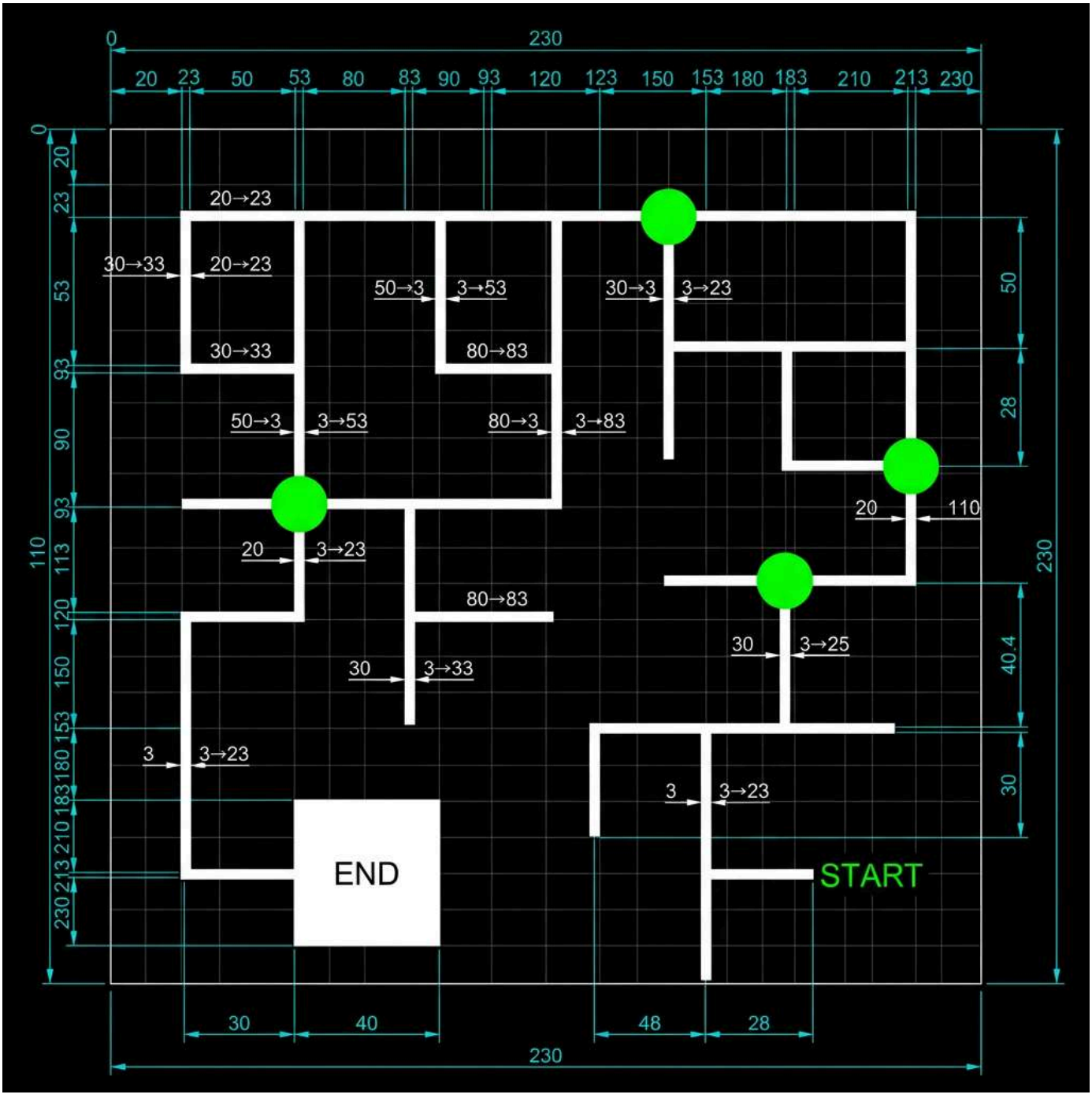


Figure 1: Arena with checkpoints

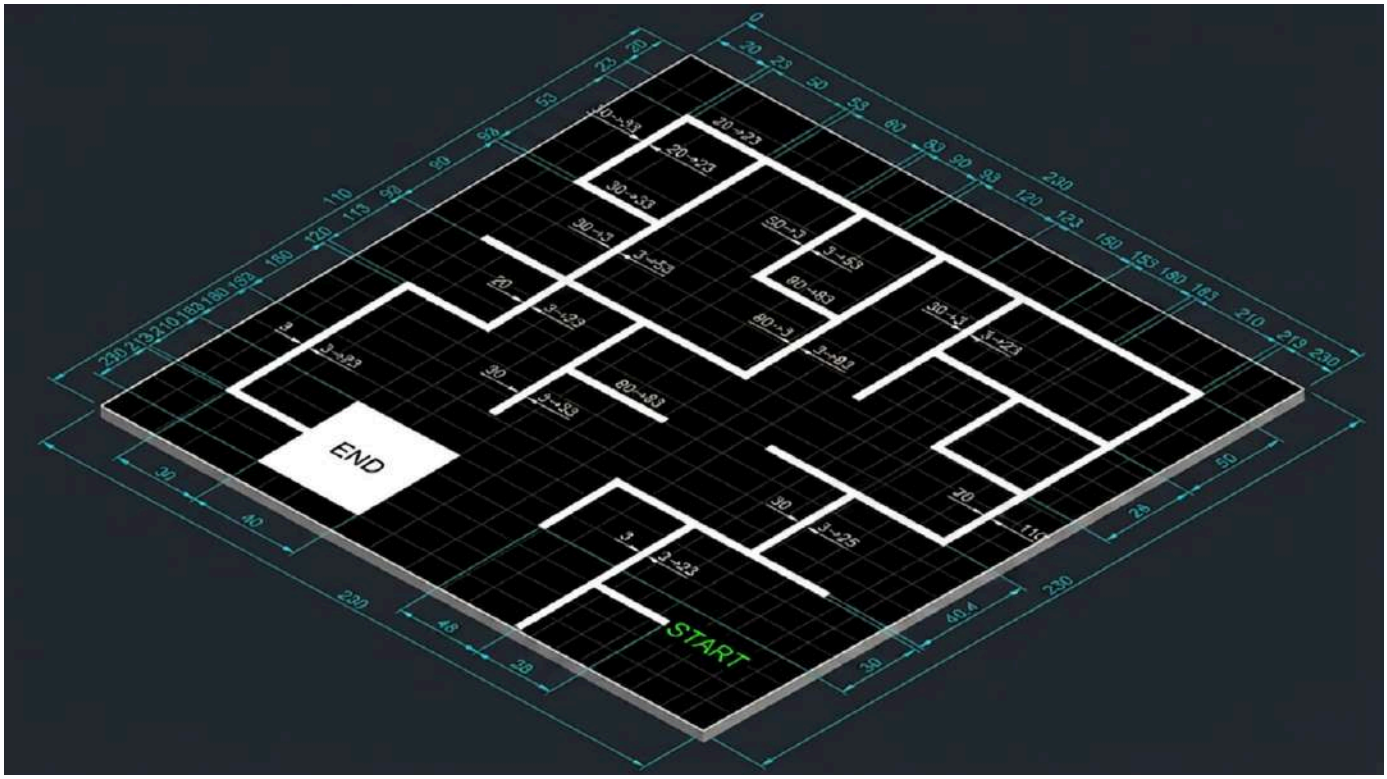


Figure 2: Isometric view

Gameplay:

The game is mainly divided into two separate bot runs.

1. The first bot run is called "dry run". In this run, the bot must start from the 'start point' and reach the 'end point(white box)' of the arena. The bot has to give a signal by glowing an LED as soon as it senses a white box below it/reach at end point. The bot has to follow an algorithm to find its path to reach 'End,' and the bot can store the turns in its memory to explore the shortest path during the second part of the journey. There are no restrictions to cover all the checkpoints.
2. The second run is an "actual run". In this run, the bot has to start from the 'start point' and end on the white box in min time by using the path stored in the bot after the first run.
3. A time period of 3 min will be provided to each team for completing the dry run, and a time period of 2 min 30 sec will be provided for the actual run.
4. Each team will be provided a total time period of 5 min 30 sec for completing both the dry run and the actual run. If a team fails to complete a dry run in 3 min, extra time will be deducted from the time period of the actual run, which is 2 min 30 sec.

Restarts:

1. The participants are allowed to take a maximum of 3 restarts in the entire match.
2. If the bot takes a restart in the first part (Dry Run) of the competition, it has to start from the checkpoint.
3. Once Dry Run is completed by the bot, the team won't be given any other chance for Dry Run.
4. If the bot takes a restart in the second part (Actual Run) of the competition, it has to start from the start zone of the arena.
5. The timer will not be set back to zero and will not be paused in any case.
6. During a restart, a contestant must not feed information about the arena to the bot. However, contestants are allowed to adjust sensors (I, position etc.) and make hardware changes.
7. The contestant must not alter the bot in any manner that reduces its weight (e.g. removal of a bulky sensor array or switching to lighter batteries to get better speed). The organisers reserve the right to arbitrate in such circumstances.

Bot Specifications:

1. The bot should fit into a box of dimensions (220 mm x 220 mmx 220 mm).
2. A Bot must be started by only one switch. However, a team may have an onboard switch.
3. A Bot must have a red LED that will glow once it reaches the end zone of the arena.
4. During the run, the autonomous bot must not damage the arena in any way. It is not allowed to leave anything behind or make any marks while traversing the arena. Any bot found damaging the arena will be immediately disqualified. The final decision is at the discretion of the organisers.
5. The bot must have an on-board power supply.
6. When using the electric power supply, the potential difference between any 2 points must not exceed 24 V at any point in time during the game.
7. The Machine cannot be constructed using ready-made 'Lego kits' or any ready-made mechanism. But they can make use of ready-made gear assemblies. Violating this clause will lead to the disqualification of the team.

Format of Competition:

Game Rules:

1. Teams will be given 1 minute for calibration. If any team is found to alter its code after depositing its bots, then it will be immediately disqualified from the competition. They are, however, allowed to make any other hardware changes.
2. Only one autonomous bot per team is allowed.
3. When the bot starts, no team member is allowed to touch the bot or enter the arena.
4. At the start of the task, the bot will be placed at the starting point. Only 1 team member is allowed to be near the game field while starting the bot.
5. The run will start only when organisers give the signal.
6. The starting procedure of the bot should be simple and should not involve giving the bot any manual force or impulse in any direction.

General Rules

1. Only 1 member of the team is allowed to handle the bot.
2. Participants are not allowed to keep anything inside the arena other than the bot.
3. Laptops/personal computers are not allowed near the arena. Other Wi-Fi, Bluetooth, etc. devices must be switched off. The organisers reserve the right to check for these devices and their usage and disqualify the team.
4. The time measured by the organisers will be final and will be used for scoring the teams.
5. Time measured by any contestant by any other means is not acceptable for scoring.
6. In case of any disputes/discrepancies, the organisers' decision will be final and binding.
7. The organisers reserve the right to change any or all of the above rules as they deem fit. Changes in rules, if any, will be highlighted on the website and notified to the registered teams.
8. Only one team is allowed to be present during the run, other teams will have to stay outside the hall. No team is allowed to take photographs or record their run.

Judging:

1. **25 points** will be awarded if the bot crosses any of the checkpoints, but it will be counted only once for each checkpoint.
2. **30 points** will be provided if the bot successfully completes the Dry Run.
3. **30 points** will be awarded if the bot goes through the Shortest Path in the Actual Run.
4. **5 points** will be awarded if the bot lights the LED.

Scoring:

1. A = 25 points * (Number of checkpoints covered during the Dry Run).
2. B = 30 points if the bot successfully completes the Dry Run.
3. C = 180 - Total time taken in seconds to complete the Dry Run.
4. S = 30 Points if the bot successfully completes through the Shortest Path.
5. T = 150 - Total time taken to complete the Actual Run (only if the bot completes in the shortest path)
6. L = 5 points if the LED glows

7. P = Penalties
8. Total = (A + B + C + S + L + T) - P

Structure:

- Top 5 teams from each Zonal Qualifier will qualify for the Grand Finale to be held during Techfest 2026-27

Team Specifications & Eligibility:

- **Eligibility:** Open to all students with a valid ID from their educational institution.
- **Team Size:** Teams may include students from different institutes. Maximum of 4 members per team.

Certificate Policy:

1. The **Top 5** teams per zonal city advancing to Round 2 receive a **Certificate of Excellence**.
2. E-Certificates of participation will be given to the teams scoring more than the critical marks which will be decided later

Prize Money:

The Prize Money will be awarded to the top 3 winners of the final round via NEFT and will be processed by May 2027.

Prize Money Submission Process

- **Submission Channel:** Complete the prize money form **directly on the website** after the final round concludes. The winners would be able to view the form through their registered email ID
- **Deadline:** The form must be submitted by **December 31, 2026**.

Details Needed for the Website Form

When filling out the form on the website, make sure you have the following bank account details ready:

1. Account Holder's Name
2. Account Number
3. Bank Name and Branch Name
4. IFSC Code